

GUARDIAN: The AI Socratic Teaching Assistant

About GUARDIAN
GUARDIAN ("Guardrailed Understanding and Reasoning in Digital Instruction for Academic Novices") is an AI-powered programming tutor built as part of an MSc dissertation at the West University of Timișoara. Unlike traditional AI assistants that give you the answer directly, GUARDIAN uses the Socratic method which guides you toward the solution through hints and questions, helping you build problem-solving skills rather than copy-paste habits. The platform supports multiple LLM providers (Gemini, ChatGPT, Claude), features a built-in code editor, and includes tools like automatic code analysis, course material integration, and interactive code-ordering puzzles.

About this study
The goal of this study is to evaluate whether Socratic AI tutoring is an effective approach to programming education, how students interact with and perceive AI-assisted learning tools, and how different LLM models compare in a tutoring context.

Procedure
The study has three steps:

- Pre-study questionnaire (this form): takes ~5 minutes, collects your background and current attitudes toward AI-assisted learning
- 1 week of platform access: you'll receive an account and can use GUARDIAN freely to practice programming exercises, get hints on your own code, or explore course materials
- Post-study questionnaire: takes ~10 minutes, collects your experience and any changes in your perceptions after using the tool

About the questionnaire
This questionnaire collects baseline information about your programming experience and attitudes toward AI-assisted learning. Your responses will only be used for research purposes.

That being said, thank you for participating in this study! 🙇🏻
The survey should not take you more than 5 minutes to complete.
Good luck! 🚀

* Indicates required question

Email *

Record **catalina.chirtes02@gmail.com** as the email to be included with my response

What best describes your current role? *

- Bachelor's student
- Master's student
- Software developer / engineer (industry)
- Other: _____

How many years of programming experience do you have? *

- Less than 1 year
- 1–2 years
- 2–3 years
- 3–5 years
- More than 5 years

What is your primary programming language? *

Your answer

How often do you use AI tools (ChatGPT, Copilot, Claude, etc.) for programming tasks? *

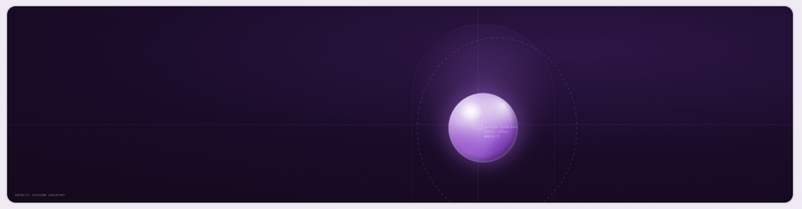
- Never
- Rarely (a few times per month)
- Sometimes (a few times per week)
- Often (daily)
- Always (every coding session)

Which AI tools do you currently use for programming? *
(Check all that apply)

- [ ] ChatGPT
- [ ] Gemini
- [ ] Claude
- [ ] GitHub Copilot
- [ ] None
- [ ] Other: _____

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Programming Self-Efficacy

Let's see how confident you are in your skills. 💻

Rate how confident you are in your ability to perform the following programming tasks. *
(1 = Not at all confident, 5 = Very confident)

	1	2	3	4	5
I can write a simple program that solves a basic problem	○	○	○	○	○
I can debug a program that has syntax errors	○	○	○	○	○
I can debug a program that has logical errors	○	○	○	○	○
I can understand code written by someone else	○	○	○	○	○
I can break a complex problem into smaller sub-problems	○	○	○	○	○
I can write a program using loops and conditional statements correctly	○	○	○	○	○
I can implement a data structure (list, dictionary, array) for a specific task	○	○	○	○	○
I can find and fix bugs in my code without someone telling me exactly what's wrong	○	○	○	○	○

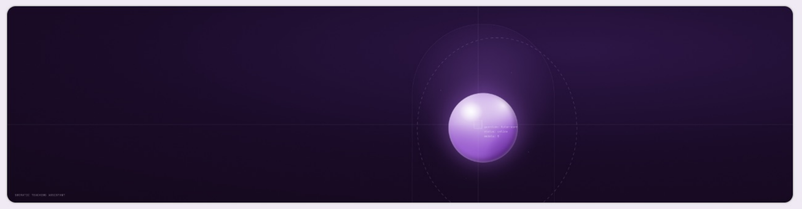
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Trust in AI for Learning

Let's see how much trust you have in LLMs. 🤖

Rate how much you agree with each statement. *
(1 = Strongly disagree, 5 = Strongly agree)

	1	2	3	4	5
I trust AI tools to give me correct programming guidance	○	○	○	○	○
I am confident that AI tutors can help me learn programming concepts	○	○	○	○	○
I can rely on AI suggestions when I'm stuck on a coding problem	○	○	○	○	○
I worry that AI tools might teach me incorrect coding practices	○	○	○	○	○
I am skeptical that AI can understand my specific programming mistakes	○	○	○	○	○
I think I learn better from human instructors than AI tools	○	○	○	○	○

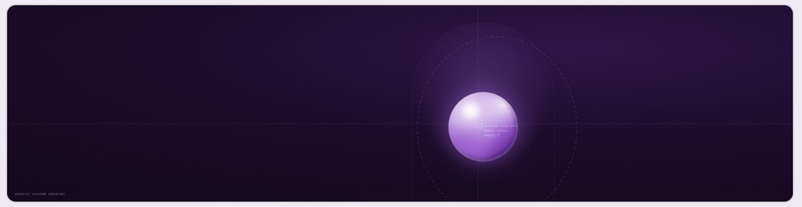
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Expectations & Consent

Let's cover these as well before you go. 🚴

What are your expectations from the GUARDIAN platform? Is there anything specific you hope to learn or experience during this study? *

Your answer

Data Consent *

Your responses will only be used for academic research purposes within this MSc dissertation. No personal data will be shared with third parties. Usage data collected by the platform (messages, feedback, session activity) is stored securely and analyzed only in aggregate for the research findings.

- ◯ I understand and agree to the use of my data as described above

Send me a copy of my responses.

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